

Metro Ticket Booking System

Requirement Analysis & Data Flow Architecture (ServiceNow Implementation)

PLATFORM

ServiceNow App Engine

DOCUMENT VERSION

v1.0 (Technical Specification)

ARTIFACT TYPE

Requirement DFD & Table Schema

Project Overview

This document details the functional requirement analysis and Data Flow Diagrams (DFDs) for building a custom **Metro Ticket Booking System** on the ServiceNow platform. The architecture leverages ServiceNow Service Catalog, Custom Tables extending task, Flow Designer, Integration Hub, and Scripted REST APIs for gate turnstile validation.

1. Primary ServiceNow Data Stores (Tables)

D1 sn_metro_station

Stores master station list, line associations, active operational status, and dynamic fare matrix records.

D3 sn_metro_payment

Audit log for financial transactions, gateway reference numbers, payment status, and refund history.

D2 sn_metro_ticket

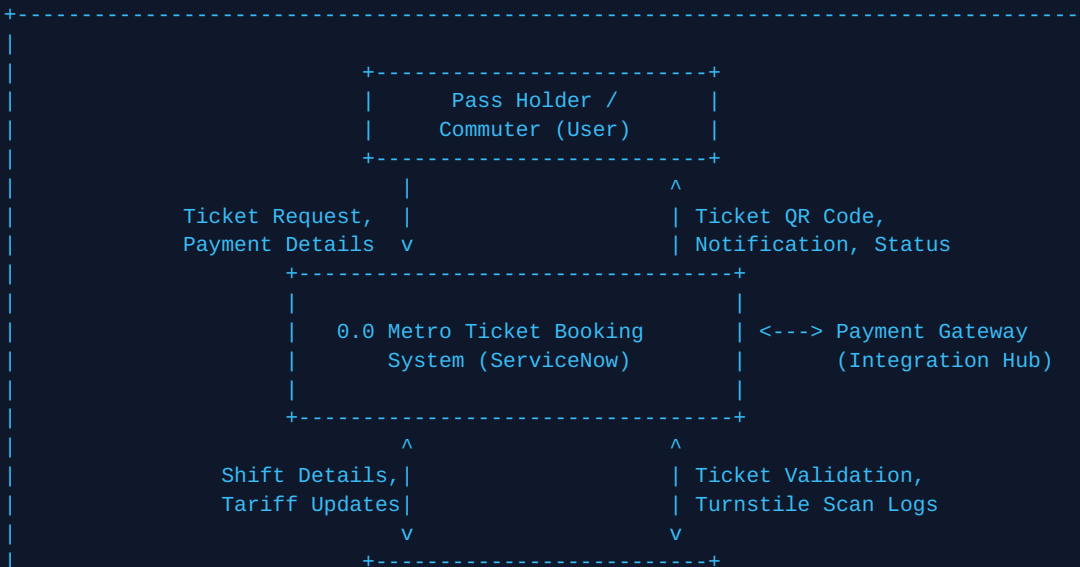
Extends task. Main operational table holding ticket IDs, QR code payloads, issue date, expiration, and lifecycle states.

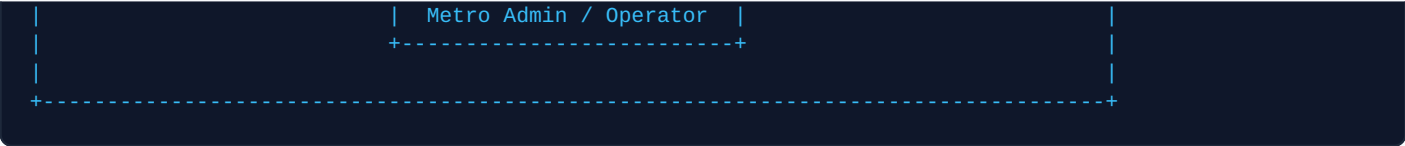
D4 sys_user

Core ServiceNow table storing commuter profiles, contact information, active passes, and notification preferences.

2. Level 0 Data Flow Diagram (Context Diagram)

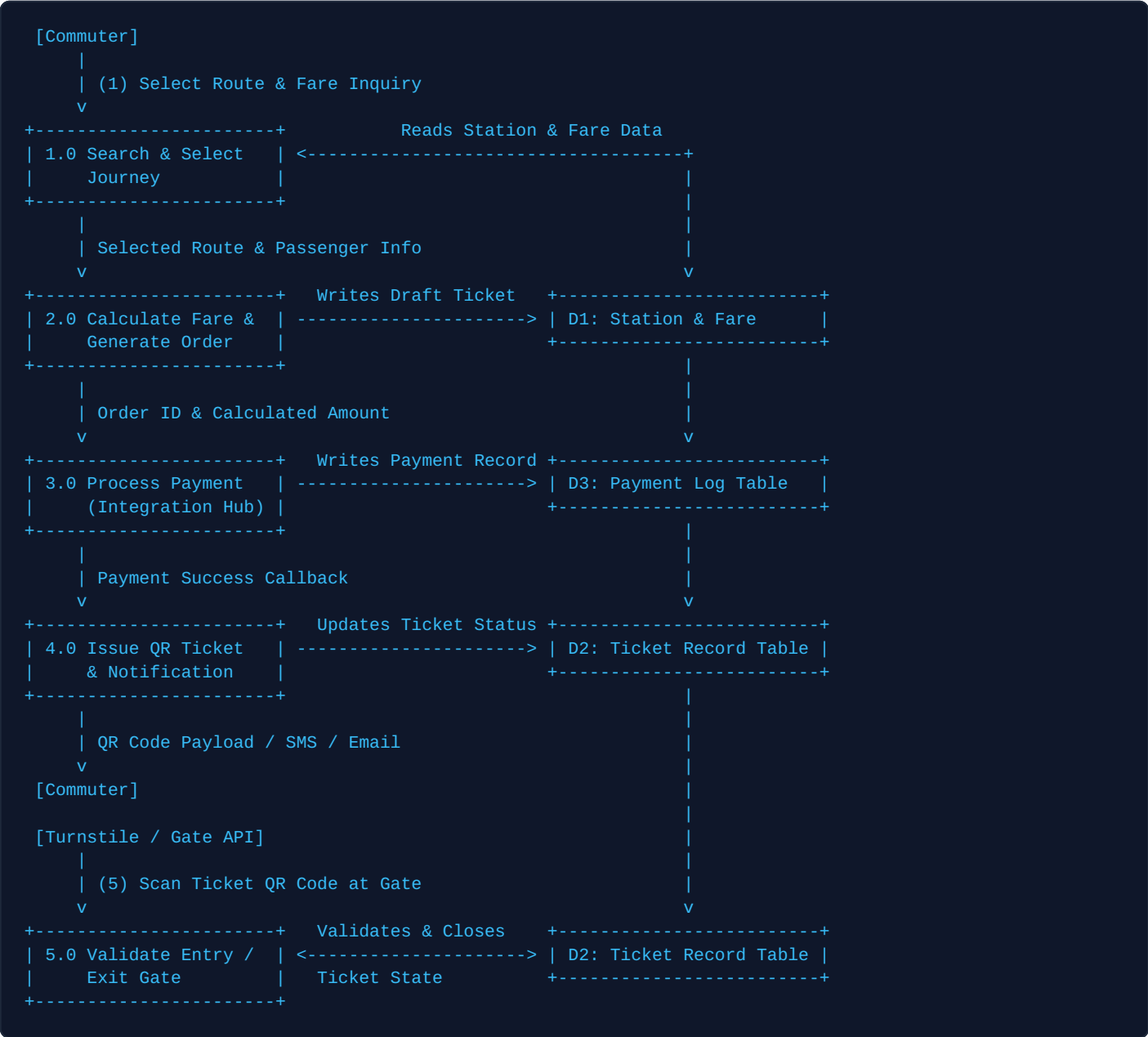
The Level 0 Context Diagram establishes the system boundaries and high-level interaction between commuters, metro operators, external payment gateways, and the ServiceNow core engine.





3. Level 1 Detailed Data Flow Diagram

Level 1 illustrates internal data transformation processes, inbound/outbound triggers, and direct interactions with ServiceNow custom tables.



4. Process Data Flow Specifications

ID	Process Name	Data Inputs	Data Outputs	ServiceNow Artifact
1.0	Search & Select Journey	Source station, destination station, ticket type	Available routes, calculated base fare	Service Portal / Mobile Widget

ID	Process Name	Data Inputs	Data Outputs	ServiceNow Artifact
2.0	Generate Order	Selected journey details, User Sys ID	Pending Ticket record (Draft status)	Script Include / Business Rule
3.0	Process Payment	Payment method, order amount, user token	Gateway transaction ID, status callback	Integration Hub REST Step
4.0	Issue QR Ticket	Payment confirmation token	Active Ticket record, QR Code attachment	Flow Designer & Notification
5.0	Validate Gate Entry	Scanned QR string from gate scanner	Gate open authorization / Denial error	Scripted REST API Endpoint

5. ServiceNow Implementation Architecture

- **Frontend / Portal Layer:** Built using Service Catalog Item & Widget in Service Portal / Now Mobile. Allows commuters to search origin/destination stations with dynamic pricing auto-calculated from `sn_metro_station`.
- **Automated Workflows:** Powered by **Flow Designer** triggered upon payment confirmation. Automatically updates `sn_metro_ticket` record to Active, encodes ticket GUID into a QR Code, and dispatches SMS / Email notifications.
- **Gate / Turnstile Integration:** Inbound **Scripted REST API** (e.g., `/api/x_metro/ticket_validation`). Turnstile scanners send HTTP POST requests with QR codes. The API verifies ticket state (Active), updates state to Used, and responds with a gate control signal within <200ms.